# A Comparative Study of Ensemble Learning Techniques for Customer Churn Prediction with Decision Threshold Optimization


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***Abstract*—Customer churn prediction is a critical task in the telecommunications industry, as retaining existing customers is significantly more cost-effective than acquiring new ones. In recent years, machine learning techniques have been widely applied to this problem, demonstrating improved predictive performance compared to traditional statistical approaches [1], [2]. This paper presents a comprehensive comparative analysis of ensemble learning methods, including voting, bagging, boosting, and stacking, for customer churn prediction. Experiments were conducted on the Telco Customer Churn dataset with extensive preprocessing and feature engineering. Multiple base learners, including Logistic Regression, Random Forest, XGBoost, and Multi-Layer Perceptron, were used to construct ensemble models.The models were evaluated using multiple performance metrics, including accuracy, F1-score, ROC-AUC, and Precision-Recall curves, to address the challenges of class imbalance [8]. Experimental results demonstrate that bagging and stacking outperform other ensemble techniques, achieving ROC-AUC scores above 0.85. Furthermore, threshold tuning significantly improves classification performance, increasing the F1-score from 0.58 to 0.64.**

**The findings highlight the importance of ensemble design, evaluation metrics, and decision threshold optimization in real-world churn prediction systems.**



***Keywords— Customer Churn Prediction, Ensemble Learning, Bagging, Stacking, Threshold Optimization, Imbalanced Data, ROC-AUC***


## I. Introduction (*Heading 1*)

Customer churn refers to the phenomenon where customers discontinue a service or switch to a competitor. In highly competitive industries such as telecommunications, customer retention plays a crucial role in maintaining profitability and long-term growth. Studies have shown that acquiring new customers is significantly more expensive than retaining existing ones, making churn prediction an essential business problem [3].

Traditional approaches to churn prediction relied on statistical methods such as logistic regression. However, these approaches often fail to capture complex, non-linear relationships present in customer data. Machine learning techniques have emerged as powerful tools for modeling such relationships and improving predictive performance [1].

Despite their effectiveness, individual machine learning models often suffer from limitations such as overfitting, high variance, or bias. Ensemble learning methods address these limitations by combining multiple models to produce more robust and accurate predictions [7]. Techniques such as bagging, boosting, voting, and stacking have been widely used in classification problems.

This paper aims to provide a comprehensive evaluation of ensemble learning techniques for customer churn prediction. Additionally, the study emphasizes the importance of handling class imbalance and optimizing decision thresholds to improve model performance.

## II. Related Work

Customer churn prediction has been extensively studied in the literature. Early works focused on traditional machine learning models such as decision trees, support vector machines, and neural networks [1], [2]. These approaches provided a baseline for churn prediction but often lacked robustness.

Ensemble learning methods have gained significant attention due to their ability to improve predictive performance. Random Forest, introduced by Breiman, is a bagging-based method that reduces variance by aggregating multiple decision trees [4]. AdaBoost, proposed by Freund and Schapire, improves weak learners by iteratively focusing on misclassified samples [5].

Stacking, introduced by Wolpert, combines predictions from multiple base learners using a meta-learner, enabling more flexible and powerful modeling [6]. Zhou provides a comprehensive overview of ensemble learning techniques and their applications [7].

Another important challenge in churn prediction is class imbalance. He and Garcia highlighted that imbalanced datasets can significantly degrade model performance if not handled properly [8]. Techniques such as SMOTE have been

proposed to address this issue by generating synthetic samples for minority classes [9].
Furthermore, Davis and Goadrich demonstrated that Precision-Recall curves are more informative than ROC curves when dealing with imbalanced datasets, as they focus on the minority class performance [10].

## III. Dataset and Data Collection

This study utilizes an updated version of the widely used IBM Telco Customer Churn dataset, specifically the latest Telco Customer Churn (11.1.3+) dataset. This dataset is an enhanced version of the original IBM dataset and includes additional attributes related to customer behavior, service usage, and billing patterns.

The dataset contains detailed information about customers in a telecommunications company and is designed for binary classification, where the objective is to predict whether a customer will churn (leave the service) or not.

### *A. Dataset Characteristics*

The dataset includes a diverse set of features categorized as follows:

- Demographic Information: Gender, senior citizen status, partner, dependents.
- Account Information**:** Tenure, contract type, billing preferences.
- Service Usage: Internet service, phone service, streaming services.
- Financial Information**:** Monthly charges, total charges, payment method

The target variable is Churn Value, where:

- 1 indicates the customer has churned
- 0 indicates the customer is retained

The dataset is inherently imbalanced, with a significantly lower proportion of churned customers compared to non-churned customers. This imbalance poses a challenge for classification models and necessitates the use of appropriate evaluation metrics.

### *B. Data Cleaning & Processing*

Several preprocessing steps were applied to ensure data quality and model compatibility:

- The "Total Charges" column was converted to a numeric format, with invalid entries treated as missing values
- Missing values were handled using median imputation to avoid bias
- Categorical variables were transformed using **one-**hot encoding
- Redundant and irrelevant features, including customer identifiers and location-based attributes, were removed

### *C. Data Leakage Prevention*

To ensure the integrity of the predictive modeling process, features directly related to churn outcomes were removed. These included:

- Churn Label
- Churn Score
- Churn Reason

Including such features would result in data leakage, artificially inflating model performance.

### *D. Train-Test Split*

The dataset was divided into training and testing sets using an **80:20 stratified split**, ensuring that the proportion of churn and non-churn instances remained consistent across both sets.

## IV. Methodology

This study proposes a comprehensive ensemble-based framework for customer churn prediction. The methodology consists of data preprocessing, base model training, ensemble construction, and threshold optimization. The overall pipeline is illustrated as a multi-stage learning process where multiple base learners are combined to improve predictive performance.

### *A. Data Preparation*

The dataset was preprocessed to ensure suitability for machine learning models. Numerical features were scaled using standardization where necessary, particularly for models such as Logistic Regression and Multi-Layer Perceptron. Categorical variables were transformed using one-hot encoding.

To prevent data leakage, all features directly related to churn outcomes (e.g., churn labels, scores, and reasons) were removed prior to training. The dataset was then divided into training and testing sets using stratified sampling to preserve class distribution.

### *B. Base Learners*

Four diverse base models were selected to capture different patterns in the data:

- **Logistic Regression (LR):** A linear model used as a baseline, effective for interpretable relationships.
- **Random Forest (RF):** An ensemble of decision trees that reduces variance through bagging [4].
- **XGBoost (XGB):** A gradient boosting algorithm that builds trees sequentially to minimize error.
- **Multi-Layer Perceptron (MLP):** A neural network capable of learning complex non-linear relationships.

The diversity among these models is crucial, as ensemble performance improves when base learners make uncorrelated errors.

### *C. Ensemble Methods*

Four ensemble strategies were implemented:

#### *a) Voting Ensemble:*

Voting combines predictions from multiple base models. Two variants were used:

- **Hard Voting:** Final prediction is based on majority class votes
- **Soft Voting:** Final prediction is based on average class probabilities

Soft voting is mathematically defined as:

$$\hat{u} = \arg\ max \sum_{i=1}^{n} p_i(y \mid x)$$

where $p_i(y \mid x)$ represents the predicted probability from model $i$.

*b) Bagging (Bootstrap Aggregation):*

Bagging reduces model variance by training multiple models on randomly sampled subsets of the training data.

$$f(x) = \frac{1}{N} \sum_{i=1}^{N} f_i(x)$$

In this study, decision trees were used as base learners for bagging. Each tree was trained on a bootstrap sample, and predictions were averaged.

*c) Boosting (AdaBoost)*

AdaBoost builds models sequentially, where each model focuses on correcting the errors of the previous one. Sample weights are updated iteratively:

$$w_i^{(t+1)} = w_i^{(t)} \cdot e^{\alpha_t I(y_i \neq h_t(x_i))}$$

This approach emphasizes difficult samples but can be sensitive to noise and class imbalance.

*d) Stacking*

Stacking combines multiple base models using a meta-learner. The outputs of base learners are used as inputs to a higher-level model:

$$y = g(h_1(x), h_2(x), \ldots, h_n(x))$$

In this study:

- Base models: LR, RF, XGB
- Meta-model: Logistic Regression

Cross-validation was used during stacking to prevent overfitting.

### D. Model Training and Validation

All models were trained on the training dataset and evaluated on a held-out test set. Additionally, stratified k-fold cross-validation was performed to ensure robustness and generalization.

Performance metrics were computed for each model, including accuracy, precision, recall, F1-score, and ROC-AUC.

### E. Threshold Optimization (Key Contribution)

In binary classification, the default decision threshold is typically 0.5. However, this threshold is often suboptimal for imbalanced datasets.

To address this, threshold tuning was performed by searching over a range of values (0.2 to 0.8). The optimal threshold was selected based on maximizing the F1-score on a validation set.

$$\hat{y} = \begin{cases} 1 & \text{if } P(y = 1 \mid x) \geq t \\ 0 & \text{otherwise} \end{cases}$$

This step significantly improved model performance, particularly in terms of recall and F1-score.

### F. Computational Considerations

To ensure efficient experimentation:

- Parallel processing was used where applicable
- A fast mode configuration was implemented for initial experimentation
- Final models were trained with higher complexity for better accuracy

## V. Result and Analysis

### A. Base Model Performance

The performance of individual base models was evaluated on the test dataset. The results are presented in Table I.

TABLE I. PERFORMANCE OF BASE MODELS

| Model | Accuracy | Precision | Recall | F1-score | ROC-AUC |
|---|---|---|---|---|---|
| Random Forest | 0.8027 | 0.6778 | 0.4893 | 0.5683 | 0.8497 |
| XGBoost | 0.7857 | 0.6078 | 0.5428 | 0.5734 | 0.8294 |
| MLP | 0.7750 | 0.6432 | 0.3422 | 0.4468 | 0.8276 |
| Logistic Regression | 0.7161 | 0.4624 | 0.4278 | 0.4444 | 0.7117 |

From Table I, Random Forest achieved the highest accuracy (0.8027) and ROC-AUC (0.8497), while XGBoost achieved the highest F1-score (0.5734). Logistic Regression showed the lowest performance, indicating that linear models are less effective for capturing complex churn patterns.

### B. Overfitting and Generalization analysis

To evaluate model generalization, training and testing performance were compared, as shown in Table II.

TABLE II. TRAIN VS TEST DIAGNOSTICS

| MODEL | TRAIN ACCURACY | TEST ACCURACY | ACCURACY GAP | TRAIN F1 | TEST F1 |
|---|---|---|---|---|---|
| XGBOOST | 0.9823 | 0.7857 | 0.1966 | 0.9665 | 0.5734 |
| RANDOM FOREST | 1.0000 | 0.8027 | 0.1973 | 1.0000 | 0.5683 |
| MLP | 0.8843 | 0.7750 | 0.1093 | 0.7341 | 0.4468 |
| LOGISTIC REGRESSION | 0.8981 | 0.7161 | 0.1820 | 0.8040 | 0.4444 |

The results indicate that tree-based models, particularly Random Forest and XGBoost, exhibit significant overfitting, as evidenced by large gaps between training and testing performance. In contrast, MLP and Logistic Regression show relatively lower overfitting but also lower predictive performance.

### C. *Test set Performance*

The performance of the proposed ensemble models was evaluated on the held-out test dataset using multiple evaluation metrics, including accuracy, precision, recall, F1-score, and ROC-AUC. The results are presented in Table III.

TABLE III. TEST SET PERFORMANCE OF ENSEMBLE MODELS

| Model | Accuracy | Precision | Recall | F1-score | ROC-AUC |
|---|---|---|---|---|---|
| Bagging | 0.8006 | 0.6555 | 0.5241 | 0.5825 | 0.8513 |
| Stacking | 0.8041 | 0.6713 | 0.5134 | 0.5818 | 0.8502 |
| Voting (Hard) | 0.7991 | 0.6553 | 0.5134 | 0.5757 | — |
| Voting (Soft) | 0.7842 | 0.6316 | 0.4492 | 0.5250 | 0.8293 |
| AdaBoost | 0.7346 | 0.0000 | 0.0000 | 0.0000 | 0.8268 |

As shown in Table III, the bagging model achieved the highest ROC-AUC score of 0.8513, indicating superior classification capability. The stacking model achieved comparable performance with a ROC-AUC score of 0.8502. Voting-based methods showed moderate performance, while AdaBoost failed to correctly identify churn instances, resulting in an F1-score of 0.

### D. *Cross-Validation Performance*

To assess model robustness and generalization ability, stratified k-fold cross-validation was performed. The average results are summarized in Table IV.

TABLE IV. CROSS-VALIDATION PERFORMANCE

| Model | CV Accuracy | CV F1-score | CV ROC-AUC |
|---|---|---|---|
| Bagging | 0.8053 | 0.5829 | 0.8546 |
| Stacking | 0.8076 | 0.5818 | 0.8529 |
| Voting (Hard) | 0.8000 | 0.5600 | — |
| Voting (Soft) | 0.7897 | 0.5332 | 0.8282 |
| AdaBoost | 0.7346 | 0.0000 | 0.8371 |

The results indicate that bagging achieves the highest mean F1-score (0.5829), demonstrating strong stability across folds. Stacking also exhibits consistent performance, while voting methods and AdaBoost show comparatively weaker results.

### E. *Threshold Optimization*

To improve classification performance under class imbalance, threshold tuning was applied to the best-performing model (bagging). The results are presented in Table V.

TABLE V. THRESHOLD TUNING RESULTS

| Setting | Accuracy | Precision | Recall | F1-score |
|---|---|---|---|---|
| Default (0.50) | 0.8006 | 0.6555 | 0.5241 | 0.5825 |
| Tuned (0.38) | 0.7885 | 0.5826 | 0.7166 | 0.6427 |

The results demonstrate that lowering the classification threshold from 0.50 to 0.38 significantly improves the F1-score from 0.5825 to 0.6427. This improvement is primarily due to an increase in recall, indicating that more churn instances are correctly identified.

### F. *ROC and Precision-Recall Analysis*

The ROC and Precision-Recall (PR) curves for all ensemble models are illustrated in Fig. 1.

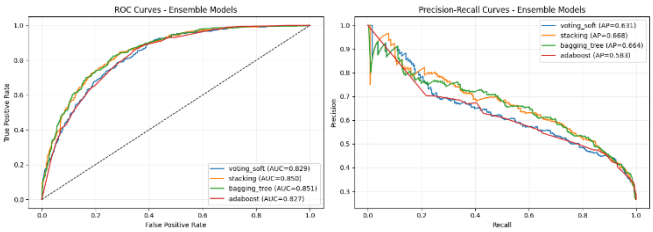


Fig. 1. ROC and Precision-Recall curves for ensemble models

The ROC curves show that bagging and stacking models provide superior class separability, achieving higher AUC values compared to other methods. However, in imbalanced classification problems, ROC curves may not fully reflect model performance.

The Precision-Recall curves provide a more informative evaluation, particularly for the minority class. The results indicate that stacking slightly outperforms bagging in maintaining higher precision across varying recall levels.

### G. *Decision of Result*

The experimental results demonstrate that ensemble learning significantly enhances customer churn prediction performance. Bagging performs best due to its ability to reduce variance, while stacking benefits from combining multiple diverse models.

AdaBoost underperforms due to its sensitivity to noise and class imbalance, leading to poor recall and F1-score. Voting methods provide stable but less optimal results.

A key finding of this study is the effectiveness of threshold tuning. Adjusting the decision threshold leads to a substantial improvement in recall and overall F1-score, making it highly suitable for real-world churn prediction scenarios where identifying potential churners is critical.

## VI. Discussion

The experimental results provide several important insights into the effectiveness of ensemble learning techniques for customer churn prediction.

First, the results demonstrate that **ensemble methods significantly outperform individual models** in terms of predictive performance and robustness. Among the evaluated approaches, the bagging model achieved the highest ROC-AUC score and the most stable cross-validation performance. This can be attributed to its ability to reduce variance by

aggregating multiple decision trees trained on bootstrapped samples. The diversity among these trees allows the model to generalize well to unseen data.

The stacking model also performed competitively, achieving performance comparable to bagging. This indicates that combining heterogeneous base learners using a meta-classifier is an effective strategy. The ability of stacking to leverage complementary strengths of different models contributes to its strong performance, particularly in terms of precision-recall characteristics.

In contrast, AdaBoost showed significantly poorer performance, failing to correctly classify churn instances. This behavior can be explained by its sensitivity to noisy data and class imbalance. Since churn datasets typically contain fewer positive instances, the iterative reweighting mechanism of boosting can overemphasize difficult or noisy samples, leading to degraded performance.

Voting-based methods provided moderate results. While they benefit from model diversity, they lack the adaptive learning capability of stacking and the variance reduction strength of bagging. As a result, their performance remains inferior to more advanced ensemble strategies.

A critical observation from this study is the impact of class imbalance on model performance. Although accuracy values remain relatively high across models, the F1-score and recall provide a more realistic assessment. This highlights the importance of selecting appropriate evaluation metrics when dealing with imbalanced datasets.

One of the most significant contributions of this work is the application of **threshold optimization**. By adjusting the classification threshold from 0.50 to 0.38, the model achieved a substantial improvement in F1-score. This improvement was primarily driven by an increase in recall, enabling the model to correctly identify a larger proportion of churn customers. In real-world scenarios, this is particularly valuable, as failing to identify potential churners can result in significant revenue loss.

Furthermore, the comparison between ROC and Precision-Recall curves provides additional insights. While ROC curves suggest strong overall performance, Precision-Recall curves reveal differences in minority class handling. This supports the argument that PR curves are more informative for imbalanced classification problems.

Despite the strong performance of the proposed approach, several limitations remain. The study does not incorporate advanced imbalance handling techniques such as SMOTE or cost-sensitive learning, which could further improve performance. Additionally, deep learning-based approaches were not explored and may provide further gains in predictive accuracy.

Overall, the findings confirm that ensemble learning, combined with proper evaluation strategies and threshold tuning, provides an effective solution for customer churn prediction.

## VII. Conclusion and Outcome

This paper presented a comprehensive study on customer churn prediction using multiple ensemble learning techniques. The objective was to evaluate the effectiveness of different ensemble strategies, including bagging, boosting, voting, and stacking, in handling imbalanced churn data and improving predictive performance.

The experimental results demonstrate that ensemble learning significantly enhances model performance compared to traditional single models. Among the evaluated approaches, the bagging model achieved the best overall performance in terms of ROC-AUC and stability, while stacking produced comparable results with strong precision-recall characteristics. Voting-based methods provided moderate improvements, whereas AdaBoost showed poor performance due to its sensitivity to class imbalance and noisy data.

A key contribution of this study is the application of decision threshold optimization. By tuning the classification threshold, the model achieved a substantial improvement in F1-score, primarily driven by increased recall. This highlights the importance of post-processing techniques in improving model effectiveness, particularly in real-world scenarios where identifying churn customers is critical.

Furthermore, the study emphasizes the importance of using appropriate evaluation metrics for imbalanced datasets. While accuracy alone may provide misleading insights, metrics such as F1-score and Precision-Recall curves offer a more reliable assessment of model performance.

Despite the promising results, this work has certain limitations. The study does not incorporate advanced imbalance handling techniques such as synthetic sampling or cost-sensitive learning. Additionally, more complex deep learning architectures were not explored.

Future work will focus on integrating dynamic ensemble selection, advanced imbalance handling methods, and deep learning-based approaches to further improve churn prediction performance.

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